



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 2

Introduction to Programming

SESSION / ASSIGNMENT – 14

TASKS :

1. Given the following buggy code meant to calculate the total price of a Zomato order, identify and fix the syntax and runtime errors:

```
let items = ["Burger", "Pizza", "Fries"];
```

```
let prices = [120, 250, 90];
```

```
let total = 0;
```

```
for (i = 0; i < items.length; i++) {
```

```
    total += prices[i]
```

```
}
```

```
console.log("Total price is: " + total);
```

Hint: Watch for assignment and loop variable issues.



```
#include <stdio.h>

int main(){
char items[3][20] = {"Burger", "Pizza", "Fries"};
int prices[3] = {120, 250, 90};
int total = 0;
int i;

    for (i = 0; i < 3; i++){
        total += prices[i];
    }
printf("Total price is: %d",total);
}
```

OUTPUT:

```
Total price is: 460
```

2. Rewrite the following code to improve its indentation and add comments explaining each step, so that a beginner can understand what it does:

```
function isEven(num){if(num%2==0){return true;}else{return false;}}
```



```
#include <stdio.h>

int isEven(int num)
{
    // Check if the number is divisible by 2
    if(num % 2 == 0)
    {
        return 1;
    }
    else
    {
        return 0;
    }
}

int main()
{
    int number = 7;

    if(isEven(number))
    {
        printf("Even Number\n");
    }
    else
    {
        printf("Odd Number\n");
    }

    return 0;
}
```

OUTPUT:

If number is even

```
Even Number
```

If number is odd

```
Odd Number
```

3. Write a function `formatFollowersCount(count)` that takes a number and returns a formatted string like Instagram: 1500 as '1.5K', 1200000 as '1.2M', and numbers below 1000 as-is. Add clear comments and use proper indentation.



```
#include <stdio.h>

void formatFollowersCount(int count)
{
    // Check if followers are in millions
    if(count >= 1000000)
    {
        printf("%.1fM\n", count / 1000000.0);
    }
    // Check if followers are in thousands
    else if(count >= 1000)
    {
        printf("%.1fK\n", count / 1000.0);
    }
    // Print the original number if below 1000
    else
    {
        printf("%d\n", count);
    }
}

int main(){

    formatFollowersCount(1500);
    formatFollowersCount(1200000);
    formatFollowersCount(850);

    return 0;
}
```

OUTPUT:

```
1.5K
1.2M
850
```

4. You are given a piece of JavaScript code that should print all even numbers from 1 to 10, but it doesn't work as expected:

```
for (let i = 1; i <= 10; i++) {  
  if (i % 2 = 0) {  
    console.log(i)  
  }  
}
```

Find and fix the error, then rewrite the code with good indentation and at least one comment explaining the logic.



```
#include <stdio.h>  
  
int main()  
{  
    int i;  
  
    // Loop through numbers from 1 to 10  
    for(i = 1; i <= 10; i++)  
    {  
        // Print only even numbers  
        if(i % 2 == 0)  
        {  
            printf("%d\n", i);  
        }  
    }  
  
    return 0;  
}
```

OUTPUT:

```
2  
4  
6  
8  
10
```